

Nate O'Farrell

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SUMMARY

Director of Infrastructure and Platform Engineering operating at the depth of a Principal Engineer. 15+ years architecting distributed systems and delivering production-critical infrastructure across cloud, on-prem, and hybrid environments, scaling organizations from 200 to 1,500+ users.

Currently leading IT, OT, and Platform Engineering at Commonwealth Fusion Systems through 1,900% growth, owning an \$8M infrastructure P&L spanning cloud, on-prem, and HPC across research, manufacturing, and enterprise. Architected and commissioned large-scale Kubernetes, multi-account AWS, HPC, and CI/CD platforms (1M+ GitHub Actions runs/month, \$5M/yr saved) at 99.99% uptime. Built the infrastructure organization from one engineer to a ten-person team; equally comfortable in the architecture review and on the keyboard.

Open to Principal/Staff IC and Director-level roles where infrastructure shapes product, the architecture problems are non-trivial, and the same person can own the architecture end-to-end, from strategy and design through hands-on implementation.

EXPERIENCE

Director of Infrastructure, OT & Platform Engineering

Commonwealth Fusion Systems

June 2023 – Present | Devens, MA

- Lead a unified IT, OT, and Platform Engineering organization, a scope normally held by three separate directors, covering SRE, DevEx, Cloud, Networking, and OT for 1,500+ employees across research, manufacturing, and enterprise.
- Grew the team from 1 to 10 engineers; established hiring plans, on-call rotations, and service catalogs to support 24/7 operations across geographically distributed sites.
- Owned an \$8M annual IT infrastructure P&L and a \$7.5M construction budget, negotiating multi-year vendor contracts and capital plans across the program portfolio.
- Directed IT/OT integration for fusion reactor R&D, aligning infrastructure with safety, regulatory, and scientific requirements while accelerating experimental iteration cycles.
- Established SRE practices and standardized on Datadog as the unified observability and on-call platform (APM, RUM, Database Monitoring, log management) across 1,000+ VMs, integrated with Prometheus, Grafana, Loki, and Tempo for OSS coverage; cut incident response time by 45% over three years.
- Built the Kubernetes platform engineering function around end-to-end GitOps (ArgoCD), Terraform-driven cluster provisioning, and Cilium/eBPF networking, supporting 30+ clusters across EKS, Rancher, BareMetal, and VMware as a self-service platform for engineering teams with policy-enforced security baselines and automated drift remediation.
- Refactored CFS's proprietary experiment telemetry platform from EC2 to Kubernetes (EKS), sustaining petabyte-scale ingest from fusion experiments without dropping a single data point during cutover.
- Operated hybrid LDAP/Active Directory across isolated factory environments and large-scale cloud HPC, sustaining >99.99% uptime through 1,900% company growth.
- Designed AWS self-hosted GitHub Actions runner architecture processing 1M+ runs/month, cutting CI spend by \$5M/yr vs. GitHub-hosted runners.
- Shipped AI/ML across infrastructure and manufacturing: Bedrock-powered internal copilots and computer-vision QC pipelines running on the factory floor.
- Drove AI-assisted development adoption across the software engineering org, rolling out Cursor Bugbot for automated PR review at an 83% merge rate; multi-year power user of Cursor and Claude Code in daily infrastructure and platform work.

Head of IT Infrastructure

Commonwealth Fusion Systems

September 2020 - June 2023 | Devens, MA

- Built Infrastructure as Code foundation across 50+ AWS accounts and 4 on-prem datacenters with 20+ reusable Terraform modules, Spacelift policies, and GitHub Actions pipelines.
- Deployed Infoblox HA DHCP and Anycast DNS across 4 datacenters, eliminating site-local DNS as a single point of failure for distributed engineering and manufacturing teams.
- Deployed Amazon EKS clusters with Karpenter for HA, auto-scaling, and zero-downtime workloads with built-in spot/on-demand cost optimization.
- Operated on-prem VMware VCF clusters, Pure Storage arrays, and TimescaleDB systems across all sites; 500 VMs handling petabyte-scale data collection and cloud offload, with SR-IOV GPU passthrough for high-performance workloads.
- Managed enterprise networking across all CFS sites (SD-WAN, BGP, Fortinet), supporting site-to-site connectivity for research, manufacturing, and corporate traffic on a single segmented backbone.
- Built HA PKI on Azure App Services supporting SCEP, OCSP, and manual cert issuance/revocation across all endpoints.
- Designed the cross-site backup and disaster-recovery architecture covering all 4 datacenters and 50+ AWS accounts, with regular restore validation and documented RTO/RPO targets per workload tier.

Enterprise Architect, Infrastructure Architecture & Security Manager

Barton Associates

October 2018 - September 2020 | Peabody, MA

- Led infrastructure and security architecture across 10 US sites for a 1,500-person regulated healthcare staffing firm, owning technical strategy, security posture, and final escalation for enterprise infrastructure.
- Directed security architecture across physical access, network, vulnerability scanning, and compliance frameworks for the regulated environment.
- Drove datacenter migrations, cloud adoption, and platform consolidation across all 10 sites, unifying identity, monitoring, and patching company-wide.
- Promoted from Senior Infrastructure Engineer after building the foundation the architecture role was created to scale.

Senior Infrastructure Engineer

Barton Associates

October 2016 - October 2018 | Peabody, MA

- Deployed 500+ servers across 10 networks supporting 2,000+ users, including system integrations for every new facility brought online.
- Led infrastructure delivery for 10+ nationwide buildouts (office, datacenter, relocations), owning rack/stack, network cutover, and day-one operational readiness.
- Coordinated 10+ vendors and contractors across \$3M in project value, delivering 100% of projects on time.
- Standardized datacenter deployment processes by developing comprehensive documentation and checklists, ensuring seamless handover to operations teams and consistent compliance across all new facilities.

PROJECTS

IDEA HPC Platform

Commonwealth Fusion Systems | github.com/cfs-energy/idea

- Primary maintainer of IDEA (Integrated Digital Engineering on AWS), the open-source HPC + virtual-desktop platform behind CFS's fusion design and simulation work.
- Architected multi-tenant infrastructure scaling to 75,000 CPU cores and multi-GPU MPI workloads concurrently, with per-team cost attribution and project-scoped IAM.
- Built spot-fleet auto-scaling and idle-shutdown automation, cutting compute costs 40% vs. fixed on-demand provisioning.

MFG 205

AWS re:Invent | youtube.com/watch?v=vrA-KiYXTug

- Co-presented session MFG 205 at AWS re:Invent 2022 on cloud HPC for fusion design and simulation; 500+ live attendees, recording on AWS's YouTube channel.

Sleepbar

Indie macOS App | sleepbar.app

- Native macOS 14+ menu bar app for scheduling sleep at a specific time or after a duration; built end-to-end in Swift/SwiftUI.
- Solo end-to-end build outside day job: product, design, Swift/SwiftUI implementation, marketing site, and release pipeline, shipping a polished native app at indie scope.

EDUCATION

Drexel University

Information Technology | 2008–2010

CERTIFICATIONS

AWS Solutions Architect Professional

2018, 2021

SKILLS

Cloud & Platform: AWS (EC2, RDS, S3, Lambda, CloudFront, ECS, EKS, Bedrock), multi-account Organizations, CloudFormation, CloudWatch, GCP, GKE

Container & Orchestration: Kubernetes (EKS, Rancher), Helm, Docker, Karpenter, Cilium, eBPF

IaC & DevEx: Terraform, Ansible, Spacelift, ArgoCD, GitOps, GitHub Actions (self-hosted runners at 1M+ runs/month)

AI/ML: AWS Bedrock, computer vision (manufacturing QC), LLM application integration, GPU workload orchestration, AI-assisted development (Cursor, Claude Code, Cursor Bugbot for PR review)

Networking: SD-WAN, BGP, Fortinet, Juniper, MPLS, IPSec, Infoblox DNS/DHCP

Data & Storage: petabyte-scale ingest pipelines, Pure Storage, TimescaleDB, PostgreSQL, Aurora, DynamoDB, S3 data lakes

Virtualization & On-Prem: VMware vSphere/VCF, SR-IOV GPU passthrough, BareMetal, hybrid cloud

Observability: Datadog (APM, RUM, Database Monitoring, on-call, log management), Prometheus, Grafana, Loki, Tempo, LogicMonitor, Elastic Stack

Identity & Security: Okta, LDAP/Active Directory, PKI/SCEP/OCSP, SIEM, Nessus, security architecture

Programming: Python, Go, JavaScript/Node.js, Bash, PowerShell